Joint phasing of sequence variants and DNA methylation from long reads with LongPhase 2

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Abstract

Long-read sequencing captures SNVs, indels, structural variants (SVs) and 5-methylcytosine (5mC) on the same molecules, yet phasers use these classes separately. Here we present LongPhase 2, which phases all four in one graph and uses a graph neural network to withhold likely misphased variants. Scored against the telomere-to-telomere (T2T) HG002 assembly on nanopore and PacBio HiFi data at 10–60× coverage, LongPhase 2 had 1.7- to 5.0-fold lower SNV switch error rates than widely used read-based phasers and 1.5- to 5.4-fold lower rates than methylation-aware phasers at every coverage, and phased SVs with 1.3- to 2.3-fold lower error rates than WhatsHap. Co-phasing all four classes lengthened phase blocks by 18–46% while switch error rates stayed at least 2.5-fold below those of WhatsHap. Switch errors shared by all phasers under the earlier GIAB benchmark largely disappeared under the T2T-derived benchmark, which distinguished the phasers more clearly, particularly in segmental duplications and other difficult regions. LongPhase 2 runs up to 8.9-fold faster on one thread and, as a drop-in replacement for LongPhase, brings joint genetic and epigenetic phasing to established long-read workflows.

Introduction

Haplotype phasing, the assignment of heterozygous variants to their parental chromosomes, is essential for interpreting the diploid human genome. Determining whether two variants lie in *cis* or in *trans* is central to identifying compound heterozygosity in recessive disease[1]. Allele-specific methylation on a phased haplotype reveals its parent of origin[2]. Long-read sequencing has made read-based phasing practical at clinical and population scale. Individual reads span tens to hundreds of kilobases, link distant heterozygous sites directly and carry native base-modification signals; a single library thus yields haplotype blocks orders of magnitude longer than short reads allow[3,4]. Read-based phasing does not require parental sequencing, a population reference panel of the kind statistical phasing depends on[5], or the additional Hi-C or Strand-seq libraries used to phase whole chromosomes[6].

Read-based phasers optimize agreement between reads and candidate haplotypes through weighted minimum error correction or probabilistic models, as in WhatsHap[7], HapCUT2[8] and Margin[9]. These methods centre on single-nucleotide variants (SNVs), although WhatsHap, by

re-aligning each read to both alleles, also co-phases small and large indels with them[10]. We previously introduced LongPhase[11], which recasts phasing as a weighted allele graph, resolves haplotypes at chromosome scale and co-phases SNVs with structural variants (SVs)[12,13]. WhatsHap and LongPhase are used in several long-read analysis pipelines (Supplementary Table 6). Clair3 phases with WhatsHap by default and with LongPhase as an option, and the somatic callers ClairS and ClairS-TO use LongPhase by default[14–17]; the EPI2ME wf-human-variation workflow phases every nanopore genome with LongPhase[18]; and the SV callers Sniffles2 and Severus take as input the haplotagged alignments that either phaser writes[12,19]. Phasing errors can therefore propagate, unflagged, to downstream analyses.

Three challenges remain in read-based phasing. The first is integrating 5-methylcytosine (5mC), which long reads report alongside the sequence, into haplotype inference. Haplotype-specific CpG methylation, as at imprinted loci and sites of sequence-dependent allele-specific methylation, can link reads to a haplotype in regions lacking heterozygous sequence variants, providing additional evidence for connecting sequence-based phase blocks[20,21]. Several methylation-aware tools use that signal after sequence-based phasing. NanoMethPhase assigns methylation to haplotypes that WhatsHap has already resolved[20]; MethPhaser and HapBridge learn haplotype-specific CpGs from the reads an SNV phaser has tagged, then join adjacent blocks or flag suspected switches[21,22]; and LongHap, which co-phases SNVs, indels and SVs, adds methylation only to bridge the gaps its sequence phasing leaves[23]. Because the informative CpGs are defined by the preceding sequence phase, its errors propagate into the methylation evidence, and methylation can revise a phase only after it is built. Allele-specific methylation is moreover confined to a small fraction of CpGs, varies between cells and tissues, and is read as a per-molecule probability rather than a base[20,21]; joint modelling allows such sparse, noisy evidence to be weighed together with the sequence alleles on the same reads.

The second challenge is accounting for errors that depend on genomic context. WhatsHap weights observations in its minimum-error-correction objective, and HapCUT2 incorporates base-quality information in its likelihood model[7,8]. Base quality alone, however, cannot establish that a read originates from the locus to which it is aligned. Base-calling errors concentrate in homopolymers and tandem repeats, reads misplaced in segmental duplications create spurious linkages, and heterozygous calls in copy-number-altered regions can mislead the phasing graph[24,25]. These context-dependent errors can create switches within haplotype blocks and confound compound-heterozygosity calls and haplotype-aware somatic detection[16,19].

A third challenge is identifying unreliable phase assignments. Statistical phasers attach posterior probabilities to phased genotypes[5], and HapCUT2 uses posterior confidence to prune variants and split low-confidence blocks[8]. Such confidence estimates depend on the phasing model and may remain high when misplaced reads provide internally consistent evidence. Local sequence context and the surrounding evidence therefore provide additional information for assessing phase reliability. WhatsHap and Margin report no confidence, and LongPhase reports only simple read support, the number of a read's alleles on each haplotype and a Phred-scaled tagging quality derived from them, not the reliability of each variant's phase. Learning-based phasers, built on graph neural networks or deep reinforcement learning[26,27], also provide no per-variant reliability estimate. They learn, partly from simulated reads, to optimize the same

fragment-cut or minimum-error-correction objective as heuristic phasers and phase SNVs only, and their authors note that errors from mis-alignment in repetitive regions remain unresolved[27].

Phasers are commonly benchmarked against Genome in a Bottle (GIAB) truth sets built largely by mapping reads to GRCh38[28,29], which assess mainly the regions where mapping-based methods perform well[30]. The HG002 v4.2.1 release omits both sex chromosomes and 12% of the autosomes, mostly satellite arrays and segmental duplications[29,30], where mis-mapped reads and collapsed paralogues create false linkages[24,25]. Its phase is also partial: the original release recorded phase only for the 74% of heterozygous variants on which trio-based phasing agreed with Strand-seq and HiFi phasing by WhatsHap[29]. This consensus requirement restricted phase assessment to variants with concordant assignments across the three approaches (the phase-transferred version scored here is described in Methods). Benchmark coverage affects the detection of errors in difficult genomic regions. GIAB's growth from 85% to 92% of GRCh38 exposed eight times more false negatives in a short-read call set[29], and the complete CHM13 reference removed tens of thousands of spurious variant calls per sample[31]. The complete diploid T2T-HG002 assembly extends the available phase information[30]. It resolves both haplotypes of every chromosome, adds 701 Mb of autosomal sequence absent from v4.2.1 and phases every heterozygous site. Its projection onto GRCh38, the GIAB v5.0q benchmark[32], extends assessment into tandem repeats and homopolymers, although satellite arrays and many segmental duplications remain excluded.

Here we present LongPhase 2, which jointly phases sequence variation and allele-specific methylation, accounts for sequence context and identifies potentially unreliable phase assignments. It places 5mC sites in the same weighted phasing graph as SNVs, small indels and SVs, incorporating methylation evidence during haplotype inference. It weights every read observation by base quality and sequence context, giving less weight to homopolymers, tandem repeats and copy-number-altered regions. It scores the phasing uncertainty of every variant. A graph neural network, enabled by default, identifies isolated disagreements and the first positions of consecutive disagreement runs (phase-disagreement onsets) for selective unphasing while preserving variant genotypes. We evaluate its SNV and indel phase against both GIAB v4.2.1 and the T2T-HG002-derived benchmark, and its SV phase against the T2T-HG002 assembly. It retains the command-line interface and output formats of LongPhase.

Results

LongPhase 2 combines joint phasing with learned error detection

LongPhase 2 integrates SNVs, small indels, SVs and allele-specific 5mC sites called by its `modcall` module in a chromosome-wide phasing graph (Fig. 1a). Observations are weighted by base quality and sequence context, with filters for homopolymers, tandem repeats and candidate copy-number-altered regions (Fig. 1b). A graph neural network (GNN) examines windows centred on variants with high phasing uncertainty. The GNN predicts phase-disagreement onsets within these windows. GNN correction selectively removes phase assignments at the predicted sites, preserves variant genotypes and splits blocks disconnected by unphasing (Fig. 1c). The outputs are phased VCFs for each variant class and haplotagged alignments (Fig. 1d).

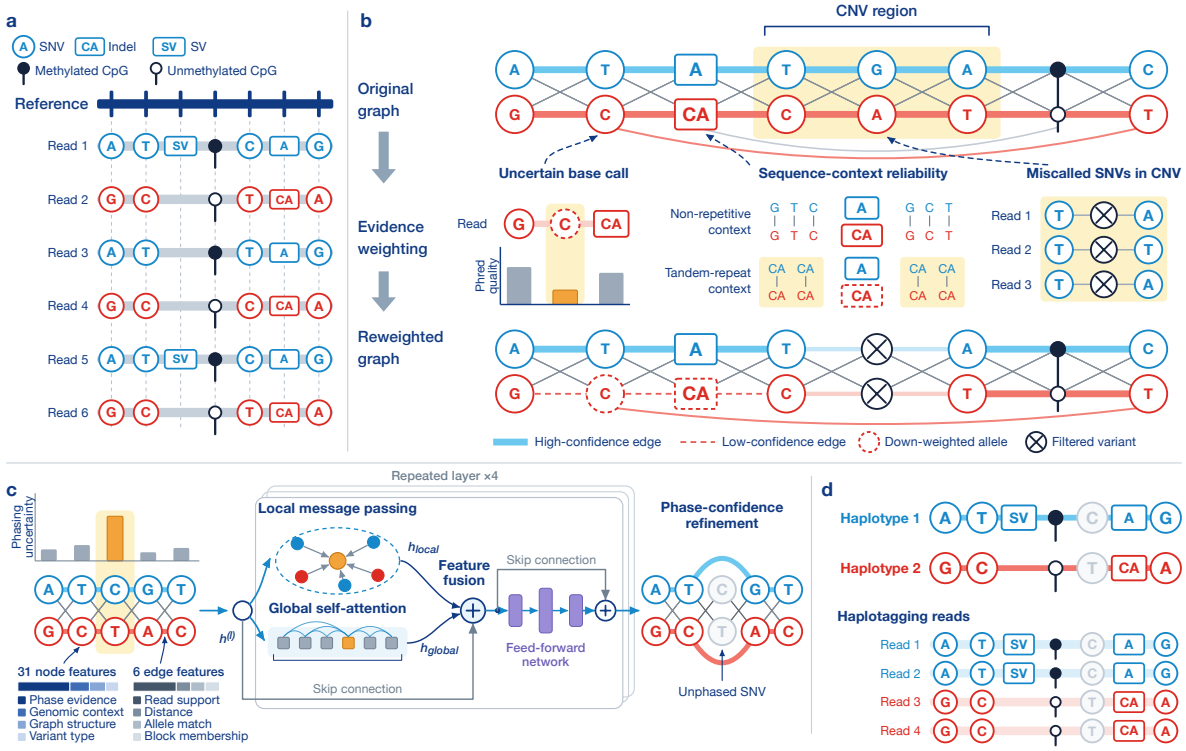


Figure 1: **Overview of LongPhase 2.** **a**, SNVs, small indels, SVs and 5mC sites carried by the same long reads enter one phasing problem; alleles are coloured by read haplotype. **b**, Weighted phasing graph. Read-supported edges link allele nodes, and each observation is weighted by base quality and sequence context; the inset shows an indel down-weighted inside a tandem repeat. **c**, A graph neural network evaluates local phasing graphs centred on variants with high phasing uncertainty (Methods). Predicted phase-disagreement onsets are selectively unphased (GNN correction; grey). **d**, Output: phased haplotypes of all four variant classes, with the selected variant left unphased (faded), and haplotagged reads.

We evaluated LongPhase 2 on HG002 nanopore R10.4.1 reads at 10–60× and PacBio HiFi reads from the same individual at 10–50× (Methods). Nanopore data comprised ten down-sampling replicates per coverage at 10–20× in 2× steps, and one each at 30, 40, 50 and 60×; HiFi and unphased-variant analyses used one replicate per coverage. All phasers received the same alignments and variant calls. We scored phase against two HG002 truth sets: GIAB v4.2.1, integrated from mapping-based call sets, and GIAB v5.0q, phased by the diploid telomere-to-telomere (T2T) assembly. Both truth sets were used in full, including variants outside their benchmark regions. SV phase was scored against the variants of the T2T-HG002 assembly, which resolves both alleles of every SV (Methods). Unless stated otherwise, results refer to SNV phasing against v5.0q and include GNN correction. Per-run values and summaries are in Supplementary Data 1 and Supplementary Tables 7–18.

LongPhase 2 reduces switch error rates across evidence types and sequencing platforms

In SNV-only phasing of nanopore data, LongPhase 2 consistently achieved lower switch error rates (0.022–0.081%) than WhatsHap (0.081–0.233%) and HapCUT2 (0.104–0.266%) across the tested coverages (Fig. 2a). These rates corresponded to 2.9–3.9-fold and 3.3–5.0-fold reductions relative to WhatsHap and HapCUT2, respectively (Supplementary Table 8). LongPhase 2 also had 1.3–1.8-fold lower block-wise Hamming distances, the fraction of misassigned variants within blocks, than WhatsHap and HapCUT2 (Supplementary Fig. 16a). When indels were included in the nanopore runs, LongPhase 2 had 3.0–3.7-fold lower SNV switch error rates than WhatsHap and 3.5–4.8-fold lower rates than HapCUT2 (Fig. 2b, solid lines). The difference was larger for consecutive variant pairs containing an indel: rates decreased from 0.100% to 0.029% for LongPhase 2 but remained at 0.75–0.91% for WhatsHap (dotted lines). The reduction increased from 8-fold at 10× to 32-fold at 60× (Supplementary Table 14). Indel Hamming distances showed the same pattern, decreasing from 6.9% to 3.6% for LongPhase 2 and remaining at 9.0–13.8% for WhatsHap (Supplementary Fig. 16b).

On PacBio HiFi data, LongPhase 2 had 1.7–1.9-fold lower SNV switch error rates than HiPhase[33], the phaser of the PacBio whole-genome workflow, 2.4–2.8-fold lower rates than WhatsHap and 2.9–3.5-fold lower rates than HapCUT2 (Fig. 2c and Supplementary Table 11). Rates decreased from 0.049% to 0.033% for LongPhase 2, from 0.085% to 0.057% for HiPhase, from 0.128% to 0.077% for WhatsHap and from 0.152% to 0.096% for HapCUT2 between 10 and 50×. The difference from WhatsHap narrowed from 2.6–2.8-fold at 10–30× to 2.4-fold at 40–50×, whereas the difference from HiPhase remained at 1.7–1.9-fold at every coverage. Hamming distances were 0.7–1.2% for all four tools; LongPhase 2 had the lowest values at 10, 30 and 40×, HiPhase at 20× and WhatsHap at 50× (Supplementary Fig. 16c). With indels, LongPhase 2 had 1.6–1.9-fold lower SNV switch error rates than HiPhase, 2.3–2.9-fold lower rates than WhatsHap and 3.1–3.5-fold lower rates than HapCUT2 (Fig. 2d and Supplementary Table 18); its rates decreased from 0.094% to 0.041%. Its Hamming distances, which include indels in these runs, were 1.3–2.4-fold lower than those of the other three tools at every coverage (Supplementary Fig. 16d).

TinkerHap[34], which is not shown in Fig. 2, also had higher SNV switch error rates than

LongPhase 2 at every coverage (replicate 1, Supplementary Table 18): 2.9–3.6-fold higher in SNV-only phasing of HiFi data, close to those of HapCUT2, 3.0–3.6-fold higher on HiFi data with indels and 3.7–5.6-fold higher on nanopore data with indels.

Lower switch error rates were accompanied by similar block contiguity (Fig. 2e–h). In SNV-only phasing of nanopore data, block N50 increased from 0.84–0.95 Mb at 10× to 2.72–2.87 Mb at 60× for LongPhase 2, WhatsHap and HapCUT2; the blocks of LongPhase 2 were up to 11% shorter than those of WhatsHap at 10–20×, because GNN correction splits blocks, and 2–5% longer at 30–60× (Supplementary Table 8). Adding indels lengthened the blocks of WhatsHap by 31–59% but increased its SNV switch error rate by 17–20%. With indels, the blocks of HapCUT2 were 1.3–1.4-fold longer than those of LongPhase 2, at 3.5–4.8-fold higher switch error rates (Fig. 2f). With indels, SVs and 5mC, the blocks of LongPhase 2 were 6–52% longer than those of WhatsHap in SNV-only mode and 6–21% shorter than those of WhatsHap with indels at 10–40× (within 1.4% at 50–60×), while its SNV switch error rates remained 2.5–3.2- and 3.0–3.8-fold lower, respectively (Supplementary Fig. 15 and Supplementary Table 10). On HiFi data, block N50 increased from 340 to 580 kb and differed by at most 3% among LongPhase 2, HiPhase, WhatsHap and HapCUT2 (Fig. 2g). With indels, the HiFi blocks of LongPhase 2 were 14–18% longer than without them, and those of HiPhase, WhatsHap and HapCUT2 were a further 1–17% longer (402–681 versus 435–778 kb; Fig. 2h); HiPhase and HapCUT2 also phased 5–9 percentage points more benchmark heterozygous indels. HiFi blocks were 2.5–4.3-fold shorter than nanopore blocks, consistent with the shorter reads. In SNV-only phasing on both platforms, multithreaded LongPhase 2 also achieved shorter wall-clock times, used less CPU time and, at all coverages but one, less peak memory than WhatsHap and HapCUT2 (Supplementary Fig. 10 and Supplementary Table 16).

LongPhase 2 phases SVs with lower error rates than WhatsHap

We scored the phase of heterozygous deletions and insertions of at least 50 bp called by Sniffles2; every tool received the same SNV and SV calls. On nanopore data, 5.8–11.3% of the SVs phased by LongPhase 2 had the wrong phase at 10–60×, compared with 11.2–18.7% for WhatsHap, 1.6–2.1-fold lower at every coverage (Fig. 3a and Supplementary Table 20). WhatsHap phased 0.7–6.3% more SVs correctly, but 1.8–2.3 times as many incorrectly, and 2.7–3.3 times as many SVs that are homozygous in the assembly. On HiFi data, the SV error rates of LongPhase 2 were 5.8–6.6%, 1.4–1.5-fold lower than those of WhatsHap (8.9–9.4%) at every coverage and similar to those of HiPhase (5.4–6.4%; Fig. 3b); HiPhase phased 7–9% more SVs correctly.

The difference depended on SV size (Fig. 3c). For SVs of at least 1 kb, LongPhase 2 had the lowest error rate at every coverage on both platforms: 8.0–12.8% on nanopore data, compared with 15.2–20.7% for WhatsHap, and 7.8–9.1% on HiFi data, compared with 10.9–13.3% for HiPhase and 13.1–14.3% for WhatsHap. For SVs of 50 bp to 1 kb on HiFi data, HiPhase had the lowest error rates (4.8–5.8%, compared with 5.6–6.5% for LongPhase 2; Supplementary Table 20). Of the deletions of at least 10 kb, across which aligners often split reads into separate alignments, LongPhase 2 phased 32–38 correctly on HiFi data with at most three errors, compared with 21–28 for HiPhase. With indels co-phased, LongPhase 2 had SV error rates of 7.2–12.1% and 7.1–7.8% on nanopore and HiFi data, compared with 16.3–22.1% and

9.9–12.4% for WhatsHap, and phased 1–9% more SVs correctly than WhatsHap on both platforms. HiPhase had higher error rates than LongPhase 2 at 10–30× and lower at 40 and 50× (6.9–8.6%), and phased 16–19% more SVs correctly (Supplementary Fig. 17).

Joint 5mC phasing gives lower error rates than methylation post-processing

On nanopore data, joint SNV and 5mC phasing by LongPhase 2 had the lowest switch error rate of the four phasers that use 5mC at every coverage (Fig. 3d and Supplementary Table 18). It had 1.5–1.8-fold lower switch error rates and 1.4–3.1-fold lower Hamming distances than MethPhaser[21] (Fig. 3d and Supplementary Fig. 16e). MethPhaser produced longer blocks, with N50 1.1-fold that of LongPhase 2 at 10× and 1.3–1.8-fold at 20–60× (0.96–5.49 versus 0.87–3.12 Mb; Fig. 3e), whereas phased fractions differed by approximately 0.2 percentage points or less (Supplementary Table 18). GNN correction accounted for 66–95% of the switch-error difference; most of the remainder came from the block joins of MethPhaser, which increased switch errors in its input by 2–15% (Supplementary Fig. 11). HapBridge[22], given the same input as MethPhaser, had 1.5–1.6-fold higher switch error rates and 1.2–1.5-fold higher Hamming distances than LongPhase 2 at every coverage, with block N50 0.95–1.34-fold that of LongPhase 2; it left its input unchanged at 10–20× and joined blocks at 30–60× while changing the number of switch errors by about 1% or less.

LongHap[23] had 2.2–5.4-fold higher switch error rates than LongPhase 2 (Fig. 3d). At 20–60×, its blocks were 1.4–2.1-fold longer than those of LongPhase 2 (Fig. 3e), but its Hamming distances were 18–20%, compared with 1.7–3.4% for LongPhase 2, consistent with long segments of reversed phase within its blocks (Supplementary Fig. 16e). At 10×, its block N50 was 0.41 versus 0.87 Mb, and it phased 60% of benchmark heterozygous SNVs, compared with 78% for LongPhase 2 (Supplementary Table 18); its Hamming distance there was lower than that of LongPhase 2 (4.4 versus 5.4%). With SNVs, indels, SVs and 5mC, LongHap had 2.4–4.5-fold higher switch error rates than LongPhase 2 and blocks 1.3–1.8-fold longer at 20–60×, with 19.5–21.9% Hamming distances (Fig. 3f and Supplementary Table 18).

Benchmark choice changes the measured separation between phasers

Against GIAB v4.2.1, differences in switch error rates among LongPhase 2, WhatsHap and HapCUT2 narrowed as coverage increased (Fig. 4a). LongPhase 2 made 1,641–1,839 switch errors across coverages, whereas WhatsHap and HapCUT2 decreased from 2,773 and 3,050 at 10× to 1,801 and 2,088 at 60×. The ratio between the highest and lowest switch error rates fell from 1.6 to 1.2–1.3 at 30–60×. When assessed against v5.0q, switch-error counts decreased with increasing coverage for all three tools, including from 1,503 to 477 for LongPhase 2, and the rate ratio increased from 3.3 to 4.3–5.0 (Fig. 4b). Relative to v4.2.1, v5.0q reduced errors for LongPhase 2 but increased them for HapCUT2 at every coverage and for WhatsHap below 60×. Hamming distance changed little (Supplementary Table 9).

Coincident switch errors accounted for much of the convergence under v4.2.1. The fraction of LongPhase 2 switch errors shared by both other tools increased from 63% at 10× to 93% at 60×. At 60×, these errors accounted for 86% of WhatsHap errors and 74% of HapCUT2 errors; corresponding shares under v5.0q were only 38%, 10% and 8% (Fig. 4c). In shared benchmark

regions at 30–60×, each tool made 1,133–1,492 switch errors against v4.2.1 but 32–393 against v5.0q (Supplementary Table 13). The truth sets thus differ in phase at many shared sites, although possible v4.2.1 phase-transfer errors have not been checked in the reads.

The separation between phasers also depended on genomic context. Within the v5.0q benchmark regions, LongPhase 2 had the lowest switch error rate of the three tools at every coverage. Rates were 2.1-fold lower than for WhatsHap and 2.6-fold lower than for HapCUT2 at 10×, increasing to 3.3–3.8-fold and 7.8–9.5-fold lower at 30–60× (Supplementary Table 13). Among GIAB genomic strata[25], the separation was largest in segmental duplications (3.2–5.4-fold at 10 and 60×) and smallest outside difficult regions (1.7–2.2-fold; Fig. 4d,e). The relative advantage of LongPhase 2 was greater in regions covered by v5.0q but not v4.2.1[32]: WhatsHap and HapCUT2 had 4.2–8.3 and 7.1–11 times its switch error rate, compared with 1.7–2.5 and 2.0–9.4 times in regions covered by both benchmarks (Fig. 4d,e). These comparisons involved 18–41 switch errors for LongPhase 2 in the newly covered regions per coverage.

Outside the benchmark regions of both truth sets, where the truth phase is least certain, switch errors accounted for an increasing share of each tool's total as coverage rose. This share increased from 28% to 69% for LongPhase 2 and from 43–46% to 62–75% for the other tools between 10 and 60×. Rates in these regions were 0.8–1.1% for LongPhase 2 and 2.6–4.9% for the other tools. These sites contributed 50–56% of the genome-wide difference in switch-error counts at 10× and 60–77% at 60×. We therefore treat this component of the comparison as exploratory (Supplementary Table 13).

LongPhase 2 phases slightly fewer benchmark SNVs and places fewer genotype errors on haplotypes than WhatsHap

LongPhase 2 phased slightly fewer benchmark heterozygous SNVs than WhatsHap and HapCUT2 despite identical input calls. We measured completeness as the fraction of benchmark heterozygous SNVs phased. WhatsHap and HapCUT2 phased 0.7–1.0 percentage points more than LongPhase 2 at every nanopore coverage. For all three tools, the phased fraction rose steeply up to 16× and plateaued at 91–92% from 20× (Fig. 5a and Supplementary Table 8). Most of the difference preceded GNN correction (0.6–0.9 percentage points); correction added 0.1–0.3. On HiFi data, WhatsHap and HapCUT2 phased 0.4–0.5 percentage points more and HiPhase 0.2–0.3 percentage points more, and all four tools increased from 87% at 10× to 92% at 20–50× (Fig. 5c). With indels, WhatsHap phased 2.5–4.5 percentage points more benchmark heterozygous indels, with a larger difference at higher coverage (Supplementary Table 9).

A larger share of the calls phased by LongPhase 2 matched benchmark heterozygous SNVs. Among heterozygous SNV calls phased on chr1–22, the share matching a benchmark heterozygous SNV increased from 89.6% to 93.3% for LongPhase 2 and from 86.2% to 91.5% for WhatsHap (Fig. 5b). The share was higher for LongPhase 2 in every replicate, although the difference narrowed from 3.4 percentage points at 10× to 1.8 at 50–60×. F1, the harmonic mean of this share and completeness, was also higher in every replicate. The difference decreased from 0.9–1.1 percentage points at 10–18× to 0.4 at 50–60× (83.3–92.3% for LongPhase 2 and 82.3–91.9% for WhatsHap; Supplementary Table 8). On HiFi data, the share was also highest for LongPhase 2 at every coverage, increasing from 92.2% to 95.0%, compared with 90.3% to 94.6% for WhatsHap

and 91.1% to 93.9% for HiPhase (Fig. 5d and Supplementary Table 11). The difference from WhatsHap narrowed from 1.8 percentage points at 10–20× to 0.3 at 50×, whereas the difference from HiPhase remained at 1.1–1.5 points.

LongPhase 2 also phased fewer genotype errors than WhatsHap. These errors place an incorrect allele on a haplotype when a homozygous site is called and phased as heterozygous[32]. At positions with a homozygous v5.0q genotype on chr1–22, WhatsHap phased 27,135, 13,706 and 11,815 heterozygous calls at 10, 30 and 60×, compared with 14,154, 9,578 and 8,339 for LongPhase 2 (1.4–1.9 times as many; Fig. 5e). WhatsHap also phased 1.3–1.4 times as many calls with a different allele in v5.0q or no v5.0q record. The latter category included 190,344 calls for WhatsHap and 147,307 for LongPhase 2 at 60×; benchmark absence alone does not establish a genotype error.

Calls phased by only one tool explained these differences (Supplementary Table 17). WhatsHap phased 124,000–151,000 calls that LongPhase 2 left unphased; 15–20% matched benchmark heterozygous SNVs. Among benchmark-matching calls in blocks with at least two assessed variants, 22.8%, 16.2% and 9.9% were phased incorrectly at 10, 30 and 60×, three to six times its overall rate. Conversely, LongPhase 2 phased 34,000–50,000 calls that WhatsHap left unphased. About 90% were absent from the benchmark, and 92–93% lay in segmental duplications or outside one-to-one assembly alignments. Among the 958–1,489 benchmark-matching SNVs assessed in such blocks, 12.7–14.9% were phased incorrectly. Calls phased by either tool alone also overlapped that tool's switch-error intervals 2.4–9.9 times more often than its phased calls overall.

At higher sequencing coverage, calls left unphased by LongPhase 2 alone included more high-depth sites than those left unphased by WhatsHap alone (Supplementary Fig. 12). At 10×, 62–71% of calls in each set had depth below 20×. At 30 and 60×, this remained true for 51–59% of calls left unphased by WhatsHap alone, compared with 12–17% of those left unphased by LongPhase 2 alone. The latter set had a depth peak near the sequencing coverage and included more high-depth calls. At 60×, 28% had depth of at least 90×, and 15.5% reached the variant caller's depth cap. Only 3% of calls left unphased by WhatsHap alone reached 90×.

The excess depth persisted when measured relative to sequencing coverage. At 30 and 60×, 28% and 17% of calls left unphased by LongPhase 2 alone had at least twice the sequencing coverage, compared with 8% and 2% for WhatsHap alone. Such depth can arise when reads from two paralogous copies map to one locus, a known source of false variant calls[32]. At 60×, 3,669 calls left unphased by LongPhase 2 alone had depth of at least 90× within v5.0q benchmark regions; 92.8% were absent from the benchmark VCF and thus classified as false positives by this benchmark. However, calls left unphased by LongPhase 2 alone overlapped segmental duplications or regions outside one-to-one assembly alignments less often than those left unphased by WhatsHap alone (54–75% versus 92–93%). The source of the excess depth remains unresolved.

At 60×, SNVs unphased by GNN correction lie mostly outside one-to-one assembly alignments

Most SNVs unphased by GNN correction were absent by position from the v5.0q VCF. GNN correction unphased 61,389–64,201 SNVs per genome at 10–60× (one replicate per coverage). This step accounted for 17–20% of the calls left unphased by LongPhase 2 alone at 10, 30 and 60×; the rest were withheld before correction. Among GNN-unphased SNVs, the share absent by position from the v5.0q VCF increased from 84.0% to 93.6% with coverage, while the heterozygous share decreased from 10.9% (6,684) to 3.5% (2,156); the remainder were homozygous (Fig. 5f). Allele matching reduced the heterozygous counts to 6,409 and 1,999. Benchmark absence alone does not establish a call error because the benchmark depends on assembly alignment and variant representation.

An exploratory analysis of T2T-HG002 assembly alignments placed 11.4% of the 62,230 SNVs unphased at 60× in regions where both haplotypes aligned one-to-one to GRCh38. This compared with 92.6% of a random sample phased before correction (odds ratio for lying outside these regions, 98; Fig. 5g and Supplementary Table 12). On autosomes alone, the shares were 9.1% and 92.9% (odds ratio, 131), with enrichment outside one-to-one alignments on every autosome (Fig. 5h). Sites outside the one-to-one alignment regions clustered in pericentromeric satellites, particularly at chr11:51–54 Mb, chr12:35–36 Mb and chr18:15–20 Mb. Of the 7,072 unphased SNVs within one-to-one alignments, 3,843 (54%) had no assembly variant at the same start coordinate.

This assembly comparison identifies genomic context but does not independently validate phase or allele identity. Coordinate matching can miss variants represented at another position. The background sample included sites later unphased by correction, which biases the comparison towards no difference. The one-to-one regions came from raw assembly alignments and retain assembly errors removed during benchmark curation. Moreover, v5.0q derives from the same assembly. Thus, the analysis extends the assessment beyond benchmark regions, but cannot establish whether the excluded phase assignments were correct.

GNN correction reduces genome-wide SNV switch error rates by one-third and limits indel-associated errors

Without GNN correction, LongPhase 2 had 1.9–2.5-fold lower SNV switch error rates than WhatsHap and 2.2–3.3-fold lower rates than HapCUT2 (Supplementary Fig. 13b and Supplementary Table 8). GNN correction reduced these rates by a further 32–36% at every coverage (31–38% across individual replicates at 10–20×). The associated loss of completeness decreased with coverage, from 0.26 percentage points at 10× to 0.08–0.11 at 30–60×. Block N50 decreased by 5–9% at 10–20× and 1–4% at 30–60× (Supplementary Fig. 13a,d).

The correction effect was smaller on chromosomes excluded from GNN training (chr17, chr21 and chr22) and on HiFi data. On held-out chromosomes, correction removed 22–25% of switch errors at 10–20× and 14–20% at 30–60× (9–17 per single run). It withheld 9.3 phased SNVs per removed error, compared with 8.0 genome-wide (Supplementary Table 15). Every tool had higher switch error rates on these chromosomes than genome-wide. The smaller reduction may therefore reflect both chromosome difficulty and sequence absent from training. After correction,

LongPhase 2 still had 2.7–5.1-fold lower SNV switch error rates than WhatsHap and HapCUT2 on these chromosomes across the tested coverages (Supplementary Table 15). On HiFi data, the nanopore-trained network removed 23–31% of switch errors without retraining, while reducing completeness by at most 0.12 percentage points and block N50 by 0.8–1.6% (Supplementary Fig. 13e,f).

Indels contributed most to both contiguity and error across seven combinations of variant classes, and GNN correction reduced the additional errors (Supplementary Figs. 14 and 15 and Supplementary Table 10). Before correction, adding SVs changed block N50 by at most 0.2% and adding 5mC increased it by 2–14%, with SNV switch error rates changing by at most 3%. Indels increased N50 by 29–51% (31–74% with SVs and 5mC), but increased switch error rates by 22–33%. GNN correction reduced rates by 31–37% without indels and 37–43% with indels, removing 40–71% of the additional error and 21–39% of switch errors at indel pairs (Supplementary Table 14). Block N50 decreased by 12–21% with indels and 0.3–9% without them as unphasing disconnected blocks.

After correction, each evidence class still increased contiguity (Fig. 3f, Supplementary Fig. 15e–h and Supplementary Table 10). Adding 5mC kept SNV switch error rates within 3% of SNV-only phasing and increased N50 by 2% at 10× and 8–16% from 18×. Indels increased N50 by 16–33% and switch error rates by 14–24%. Four-class phasing produced blocks 18–46% longer than SNV-only blocks (4.14 versus 2.87 Mb at 60×), with 14–27% higher SNV switch error rates. Its blocks were within 1.7% of those of SNV, indel and 5mC phasing without SVs, which were longer at every coverage except 30 and 40×, because an SV does not join two blocks on its own. Adding SVs and 5mC to indels changed indel-pair switch error rates by at most 7% (Supplementary Table 14). All configurations retained at least 2.5-fold lower SNV switch error rates than WhatsHap with or without indels.

The phased fraction varied by variant class in four-class phasing: LongPhase 2 phased 30.5–45.4% of benchmark heterozygous indels, 97.4–99.2% of the heterozygous 5mC sites called by `modcall` and 66.1–68.5% of the heterozygous SVs called by Sniffles2 (replicate 1; Supplementary Data 1).

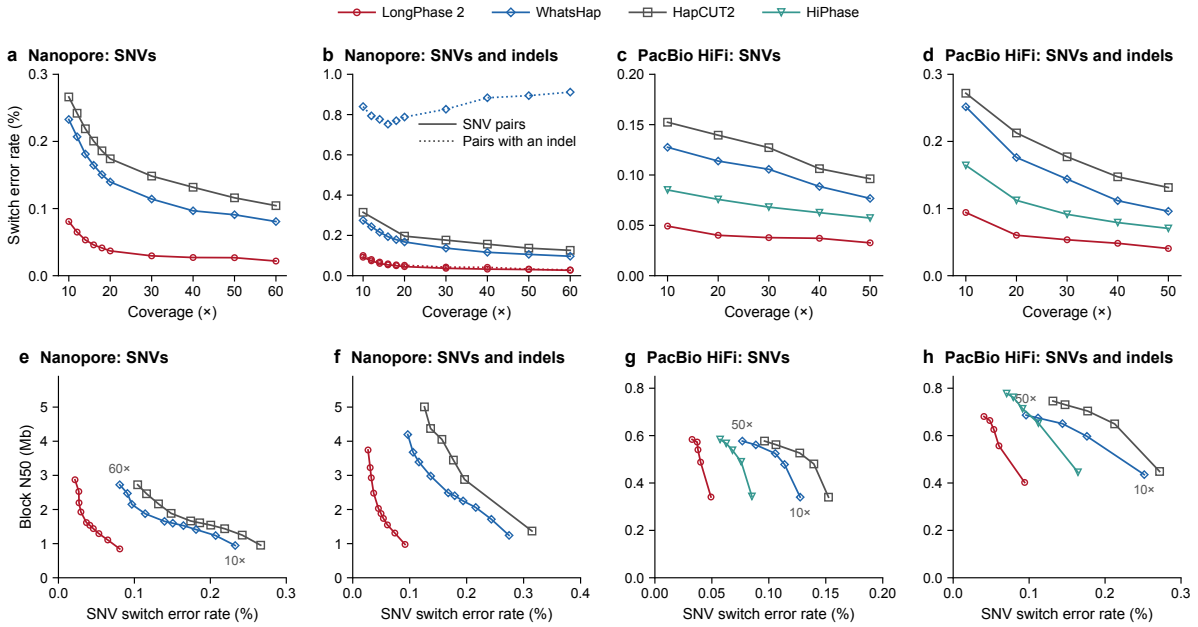


Figure 2: **Accuracy and contiguity of LongPhase 2 compared with other phasers on sequence variants from nanopore and PacBio HiFi data.** HG002 nanopore R10.4.1 reads down-sampled to 10–60× (**a**,**b**,**e**,**f**) and PacBio HiFi (Revio) reads down-sampled to 10–50× (**c**,**d**,**g**,**h**); identical variant calls for every tool, scored against the T2T-HG002-derived v5.0q benchmark. Nanopore values at 10–20× are means of ten down-sampling replicates, except for HapCUT2 with indels; all other values come from one down-sampling (Methods). Top row, SNV switch error rate against coverage; bottom row, SNV switch error rate against phase-block N50, one point per coverage from the lowest (bottom right) to the highest (top left); the N50 axes of **g** and **h** differ from those of **e** and **f**. Hamming distances are shown in Supplementary Fig. 16 and phased fractions in Fig. 5. **a**,**e**, Nanopore data: SNV phasing compared with WhatsHap and HapCUT2. **b**,**f**, Nanopore data: SNV and indel co-phasing compared with WhatsHap and HapCUT2; in **b**, solid lines score consecutive SNV pairs and dotted lines consecutive pairs that include an indel (SNV–indel or indel–indel; Supplementary Table 14). **c**,**g**, PacBio HiFi data: SNV phasing compared with WhatsHap, HapCUT2 and HiPhase; HiPhase, which has no SNV-only option, was given the calls restricted to SNVs. **d**,**h**, PacBio HiFi data: SNV and indel co-phasing compared with the same tools, scored on consecutive SNV pairs (Supplementary Table 18). On HiFi data, LongPhase 2 uses the network trained on nanopore data.

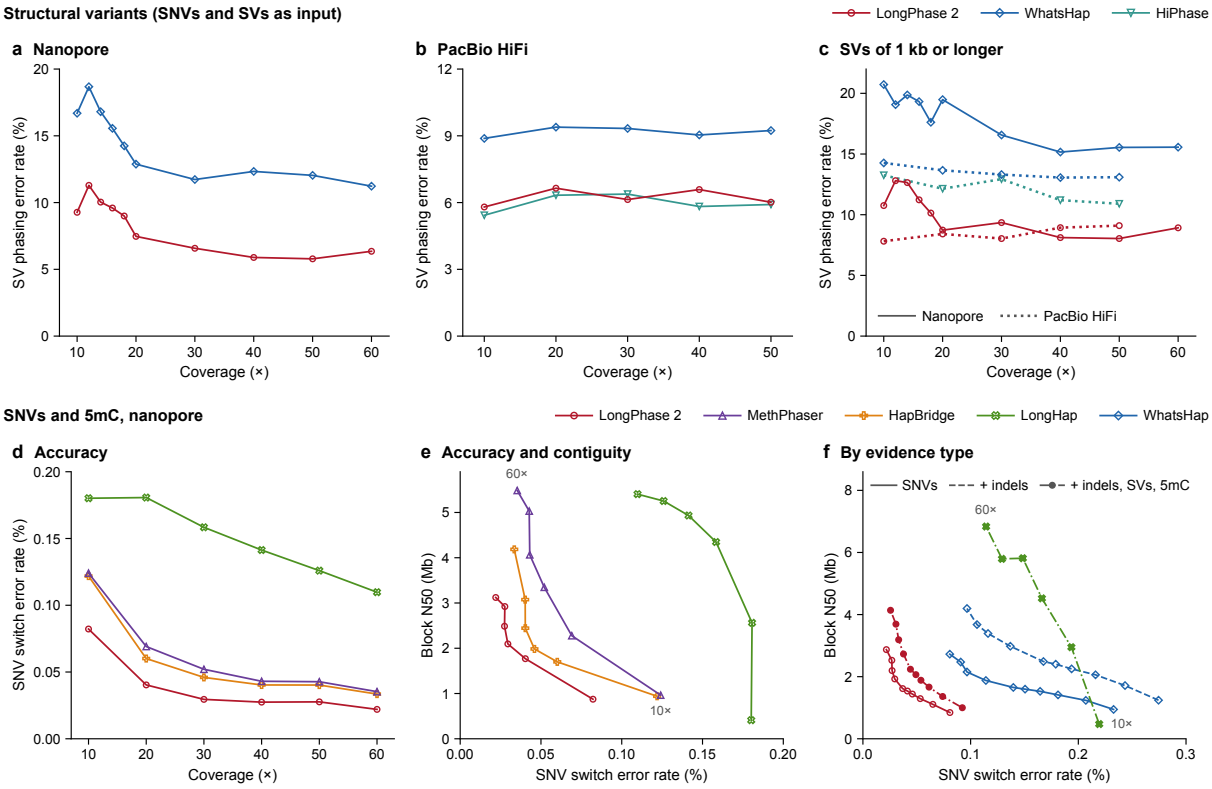


Figure 3: **Joint phasing of SVs and 5mC by LongPhase 2 compared with other phasers.** HG002 nanopore R10.4.1 reads down-sampled to 10–60× and PacBio HiFi (Revio) reads down-sampled to 10–50×, replicate 1 unless stated. SV phase was scored against the T2T-HG002 assembly and SNV phase against v5.0q. In **a**–**c**, every tool received the same SNV and SV calls (Methods). MethPhaser and HapBridge were given the SNV-only output of LongPhase 2 without GNN correction; LongHap phased the same calls with its own 5mC input (Methods and Supplementary Table 5). **a**,**b**, SV phasing error rate against coverage on nanopore (**a**) and PacBio HiFi (**b**) data, with SNVs and SVs as input (with SNVs, indels and SVs in Supplementary Fig. 17); SVs shorter than 1 kb are 90–93% of the SVs scored. **c**, SV phasing error rate of SVs of at least 1 kb on nanopore (solid) and PacBio HiFi (dotted) data. Counts and error rates by size are in Supplementary Table 20. **d**, SNV switch error rate against coverage of SNV and 5mC co-phasing on nanopore data. **e**, SNV switch error rate against phase-block N50 of the runs in **d**, one point per coverage from the lowest (bottom right) to the highest (top left). Phased fractions are in Supplementary Table 18 and Hamming distances in Supplementary Fig. 16e. **f**, SNV switch error rate against phase-block N50 by evidence type on nanopore data: LongPhase 2 with SNVs and with SNVs, indels, SVs and 5mC (means of ten replicates at 10–20×; Supplementary Table 10), compared with WhatsHap with SNVs and with SNVs and indels and with LongHap with SNVs, indels, SVs and 5mC; filled markers, four classes. Other evidence types are in Supplementary Fig. 15.

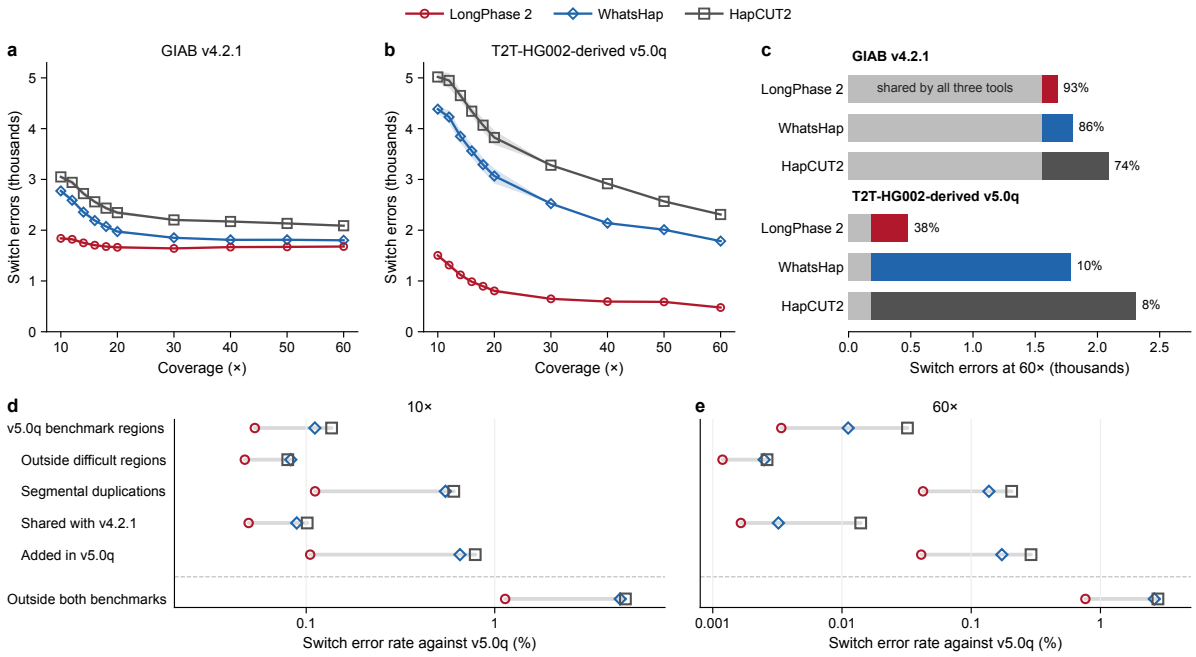


Figure 4: **The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark.** SNV-only phasing of HG002 nanopore R10.4.1 data; replicates as in Fig. 2; both truth VCFs used in full. **a**,**b**, Genome-wide switch errors under v4.2.1 (**a**) and v5.0q (**b**). **c**, Switch errors at 60×; grey, switch errors made by all three tools at the same SNV pair, with their share of each tool's total. **d**,**e**, Switch error rate against v5.0q at 10× (**d**) and 60× (**e**) inside the v5.0q benchmark regions and in subsets of them: outside difficult regions and in segmental duplications (GIAB v3.6 strata), and in the parts that v4.2.1 also covers or that v5.0q adds. Below the dashed line, outside the benchmark regions of both truth sets, where the truth phase is least certain (exploratory). A variant pair is assigned to a region when both of its variants lie in it (Supplementary Table 13).

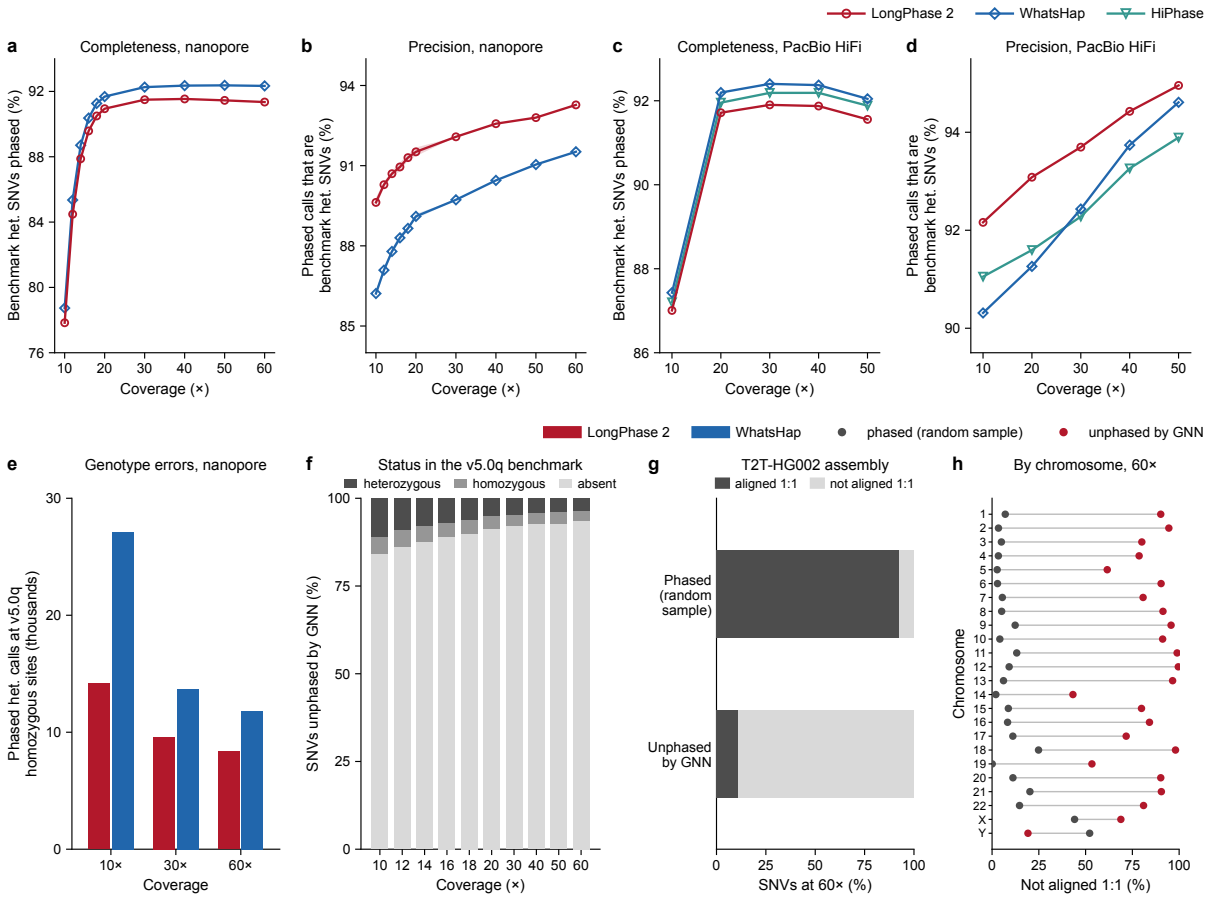


Figure 5: **Calls phased by LongPhase 2, WhatsHap and HiPhase, and SNVs unphased by GNN correction.** SNV-only phasing of HG002 nanopore R10.4.1 data (**a**, **b**, **e**–**h**) and PacBio HiFi data (**c**, **d**). **a**–**d**, coverages and replicates as in Fig. 2; **e**–**h**, replicate 1. **a**,**c**, Completeness: fraction of the heterozygous SNVs of the v5.0q benchmark phased from nanopore (**a**) and HiFi (**c**) data; every tool received the same heterozygous calls. **b**,**d**, Precision: share of the phased calls on chr1–22 that match a benchmark heterozygous SNV in position and alleles, for nanopore (**b**) and HiFi (**d**) data (Supplementary Tables 8 and 11). **c**,**d** also show HiPhase, which was given the calls restricted to SNVs. **e**, Phased heterozygous calls on chr1–22 at positions where v5.0q has a homozygous genotype (genotype errors; Supplementary Table 17). **f**, SNVs phased before and unphased after GNN correction, by their genotype in the v5.0q benchmark VCF, matched by position without a benchmark BED. **g**, At 60×, these SNVs and a random sample of SNVs phased before correction, inside or outside the regions where both haplotypes of the T2T-HG002 assembly align one-to-one to GRCh38 (exploratory). **h**, The share outside these regions for each chromosome (Supplementary Table 12).

Discussion

LongPhase 2 combines genetic variation and allele-specific methylation in a shared phasing graph with selective withholding of phase assignments predicted to be unreliable. This approach extends methods that use methylation or SVs to connect blocks in regions with few heterozygous SNVs[2,11,20,21,33]. Whereas LongHap adds methylation after sequence phasing[23], LongPhase 2 integrates both during haplotype construction. On the same reads, LongPhase 2 had 1.5–1.8-fold lower switch error rates than MethPhaser and HapBridge, which post-process an SNV phasing, and 2.2–5.4-fold lower rates than LongHap (Fig. 3d and Supplementary Table 18); for MethPhaser, most of this difference came from GNN correction. SVs placed in the same graph were phased with 1.3–2.3-fold lower error rates than by WhatsHap, and SVs of at least 1 kb with lower error rates than by HiPhase on HiFi data. The evidence classes contributed differently: methylation improved contiguity at similar SNV switch error rates, whereas indels produced most of the gains in contiguity and increases in error. The same trade-off in WhatsHap suggests that difficulties in indel representation or modelling may affect both methods. In joint SNV and indel phasing, LongPhase 2 with GNN correction had 8–32-fold lower switch error rates than WhatsHap at variant pairs containing an indel. GNN correction reduced the additional error associated with indels. More accurate indel calls could reduce the increase in phasing error associated with indel co-phasing[15,35].

Selective withholding requires a distinction between evidence agreement and confidence in phase. LongPhase 2 reports phasing entropy and weighted vote totals. Unlike model-based confidence estimates from HapCUT2[8] and statistical phasers[5], low entropy indicates agreement among available votes, not necessarily strong evidence or a low error probability. The GNN uses neighbouring graph and sequence features and may detect candidates missed by an entropy threshold alone. However, its added value over simpler withholding rules and HapCUT2's confidence pruning remains untested at matched variant retention. Its onset-prediction score is neither calibrated nor included in the output. These distinctions limit the use of the reported scores as quantitative measures of error risk.

Withholding variants and splitting blocks also change which variant pairs can be assessed. Switch error rates must therefore be considered together with completeness and contiguity. Before GNN correction, LongPhase 2 already had 1.9–2.5-fold lower switch error rates than WhatsHap at similar block N50. Most additional SNV calls phased by WhatsHap did not match a benchmark heterozygous SNV. Among assessable benchmark heterozygous SNVs phased by WhatsHap alone, 10–23% were phased incorrectly, three to six times its overall rate. Benchmark-matching SNVs phased exclusively by LongPhase 2 had a similar error rate, but were 13–26 times fewer. SVs showed the same pattern: given the same SNVs and SVs, WhatsHap phased 0.7–6.3% more SVs correctly than LongPhase 2, but 1.5–2.3 times as many incorrectly. At 60×, most SNVs unphased by the GNN lay outside one-to-one assembly alignments, where our analysis cannot establish whether their phase was correct.

Benchmark choice changed the apparent separation between methods. At 30–60×, the highest and lowest switch error rates differed by 1.2–1.3-fold against GIAB v4.2.1 but by 4.3–5.0-fold against v5.0q. Many apparent errors under v4.2.1 occurred at the same variant pairs across tools, and most disappeared when the same outputs were scored against v5.0q in shared benchmark

regions. All three phasers agreed with v5.0q at most of these sites, including WhatsHap and HapCUT2, which have no parameters trained on either truth set. Errors in v4.2.1 phase transfer are one possible explanation, but shared reads, alignments and variant calls could also contribute to correlated outputs. Independent checks with reads or parental data are needed to distinguish these explanations.

Within v5.0q benchmark regions, LongPhase 2 had 2.1- to 9.5-fold lower switch error rates than WhatsHap and HapCUT2, with the largest separation in newly covered regions. However, much of the genome-wide difference arose outside the benchmark regions of both truth sets, where the phase is least certain. This component remains exploratory. The v5.0q release evaluated variant calls rather than phase[32]. Our results support comparing phasers against assembly-phased truth sets, with phase errors reported separately inside and outside benchmark regions and stratified by genomic context[25]. Reporting which calls receive phase assignments also captures genotype errors propagated onto haplotypes, which phase-error rates alone do not measure.

Generalization remains the main evaluation limit. All data came from HG002. The network was trained on nanopore runs from the same down-sampling series, using labels from the evaluation benchmark. Genome-wide metrics include 17 training chromosomes. On held-out chromosomes, the advantage over WhatsHap and HapCUT2 persisted, but correction removed fewer errors at a higher cost in withheld SNVs. These chromosomes were excluded from GNN training but contributed to earlier tuning of the graph constants. The HiFi results without retraining support transfer between platforms within HG002, but do not establish transfer to other individuals. Evaluation in other individuals against an independent benchmark, with fixed weights and thresholds, is needed. Subsampling variability was not estimated above 20× for nanopore data or at any HiFi coverage.

Further work should test both the scope of co-phasing and its downstream effects. 5mC evaluation measured phased fractions rather than phasing accuracy, and neither SVs nor 5mC sites had supervised training labels. SV phase was scored with one SV caller against variants called from the assembly rather than a curated SV benchmark. Methylation co-phasing was assessed only on autosomes of a male lymphoblastoid cell line[21]; the method assumes diploidy and balanced allele fractions. Compatibility with pipelines that use LongPhase[14,16,18,19] enables tests of selective withholding in haplotype-aware somatic calling. More accurate germline phase could improve tumour-read assignment, but withholding phase may reduce information available to the caller. Varying the germline phaser while fixing the somatic caller would quantify this trade-off. A calibrated per-variant confidence measure could further allow downstream methods to account explicitly for uncertainty in inherited phase.

Methods

Overview of LongPhase 2

LongPhase 2 is a diploid germline phaser that takes long-read alignments, a reference genome and heterozygous variant calls, and returns phased VCFs and haplotagged alignments (Fig. 1). SNVs, small indels, structural variants (SVs) and allele-specific 5-methylcytosine (5mC) observations

are represented together in one weighted graph per chromosome. Initial haplotypes are resolved from weighted multi-neighbour votes, reads are used for a second phase-consistency pass, and a graph neural network (GNN) then scores candidate phase-disagreement onsets. Selected variants are unphased, never flipped; if this disconnects a haplotype block, the surviving components receive updated phase-set identifiers. The detailed module rules are given in Supplementary Methods 1–6. The method extends the graph formulation of LongPhase 1[11].

Input variant calls

Heterozygous SNVs and small indels were taken from PEPPER-Margin-DeepVariant[9]; homozygous and filtered records were carried through unchanged but did not enter the graph. Indels in short tandem repeats were retained as evidence but down-weighted, because a misplaced repeat indel can link the wrong alleles. SVs were taken from Sniffles2[12] or cuteSV[13] with supporting read names; homozygous SVs, duplicated positions and SVs coincident with SNVs were excluded. The exact eligibility and repeat annotation rules are in Supplementary Method 2.

Calling allele-specific methylation

Modification probabilities were read from the `MM`/`ML` tags produced by Dorado[36] and carried through alignment. A CpG was called methylated when its 5mC probability was at least 0.8, unmethylated when it was at most 0.2, and ambiguous otherwise. Forward- and reverse-strand observations were mapped to one CpG coordinate. A site was retained as a candidate allele-specific site when methylated and unmethylated reads were sufficiently balanced and the ambiguous fraction was small:

$$\frac{\min(M_R, U_R)}{\max(M_R, U_R)} \geq 0.6, \qquad \frac{A_R}{M_R + U_R + A_R} \leq 0.2, \tag{1}$$

where M_R, U_R and A_R are the three read counts. Candidates were then required to co-segregate with a neighbouring heterozygous marker. For a pair of sites, writing RR, RA, AR and AA for the four observed state combinations, linkage required

$$\frac{\max(RR + AA,\ RA + AR)}{RR + AA + RA + AR} \geq 0.9. \tag{2}$$

The same criterion was used for SNV-anchored and methylation-only linking. Only CpG dinucleotides were considered. Weak candidates were admitted in a later expansion pass, and consecutive CpGs were merged into one locus with a shared read list for both strands. The resulting methylated and unmethylated read names and depths were supplied to the phasing graph. The depth bound and expansion rules are in Supplementary Method 1.

Read-level allele observations and filtering

Alignments with mapping quality below 1, unmapped, secondary or duplicate records were excluded. SNV alleles came from the aligned base and its Phred quality; indel alleles from the CIGAR; SV alleles from a matching CIGAR event, the junction of a split alignment or the caller's read list; and 5mC alleles from the methylation read lists. Supplementary alignments

were retained as evidence, so when two alignments of the same read overlapped by more than 20% of their combined span, the shorter alignment was discarded. Under the nanopore setting, a downstream SNV was removed when two SNVs at most 2 bp apart both lay in homopolymers of length at least three. A copy-number-aware filter identified intervals bounded by excess front- or back-clipping and removed variants whose alternate-supporting reads had an unusually high mismatch load relative to reference-supporting reads. Specifically, a read's mismatch load R_m counted the heterozygous sites in the interval at which it carried the alternate allele, and each variant was scored by

$$\mathrm{MR}(v) = \frac{\bar{R}_m^{\mathrm{ALT}}}{\bar{R}_m^{\mathrm{REF}} + \bar{R}_m^{\mathrm{ALT}}}, \tag{3}$$

where the barred terms are the mean loads of reads supporting the ALT and REF alleles of v. Sites with $\mathrm{MR}(v) \geq 0.7$ were removed from all reads. The three-state interval scanner and its constants are specified in Supplementary Method 2.

Weighted phasing graph

Each heterozygous variant contributed reference and alternate allele nodes. For each read, every allele was connected to the next 35 observed variants, with no edge-distance cap. Variants followed by a gap of more than 300 kb to the next heterozygous variant were skipped during haplotype resolution, casting no votes and receiving no haplotype or entropy. A read contributed weight 1 when both observations had quality at least 12 and weight 0.1 otherwise. Indel and 5mC observations and SV alternate observations were assigned fixed quality 60; reference-allele SV observations were assigned 30. Thus all such observations have full weight at the default threshold, while reference-SV evidence is down-weighted only at a higher threshold.

For a variant pair (u, v), the four allele-pair supports were accumulated as w_{rr}, w_{ra}, w_{ar} and w_{aa}. The two possible pairings have support $P = w_{rr} + w_{aa}$ and $Q = w_{ra} + w_{ar}$, and their ambiguity is $s = \min(P, Q)/\max(P, Q)$.

Haplotype resolution by multi-neighbour voting

Variants were processed in coordinate order. Each variant cast a vote to each downstream neighbour for the pairing with larger support, except when $s > 0.7$; SNV–5mC pairs used a 0.3 threshold and required at least one read-equivalent of support. Votes had weight 1 by default, 20 for near-unanimous support ($s \leq 0.1$ with total support at least 1, or one pairing below 1 while the other was at least 1), and 0.1 when the voting variant was a tandem-repeat indel. The vote was translated using the already assigned haplotype of the upstream variant, producing totals $h_1(v)$ and $h_2(v)$. When more than three votes had support of at most 1 read-equivalent, and at least one qualifying vote remained, only votes with $s < 0.2$, weight at least 1 and a non-indel voting variant were retained. The larger total assigned the haplotype. A tie beyond the last accepted edge opened a new block; a tie within an existing block left the variant unphased and cast no votes. A variant whose votes all came from SVs, beyond the last variant reached by a vote of a small variant, also opened a new block, so that an SV was never the only link between two blocks; SVs still voted and were phased within a block. The first variant position defined the phase-set identifier. The constants were fixed by a pre-study sweep on HG002 at

10–60× and were not re-tuned for the reported experiments; the parameter tuning is described in Supplementary Method 3.

Read-based correction and phasing entropy

After the graph assignment, each read was assigned to a haplotype by a weighted majority of its phased SNV and SV alleles (weight 1), indel alleles (weight 0.1) and 5mC alleles (not used for this pass). Reads were retained only with a majority above 0.65 and total evidence above 1. For each variant, the consistency of the two read populations was

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2},$$

where n_1 counts haplotype-1 reads carrying REF plus haplotype-2 reads carrying ALT, and n_2 counts the converse. The majority pairing assigned the phase only when $\rho > 0.75$; otherwise the variant was left unphased.

For each phased variant, phasing entropy was calculated from the weighted haplotype totals after the single-read guard:

$$\mathrm{PE}(v) = -\sum_{i\in\{1,2\}} p_i \log_2 p_i, \qquad p_i = \frac{h_i}{h_1 + h_2}. \tag{4}$$

It ranges from 0 for unanimous support to 1 bit for equal support. When $h_1 + h_2 = 0$, PE was set to 0 and the case was identified from the accompanying h_1, h_2 fields; this occurs only at the first variant of a block. PE, h_1 and h_2 were written to the VCF, and accepted voting edges with their accumulated weights formed the GNN input graph.

Graph neural network for phase-error detection

Phasing entropy was used to select candidate windows for refinement. Within each window, the GNN combined phasing evidence, genomic context and graph structure to identify phase-disagreement onsets for selective unphasing, as defined by the training labels. Positive targets include isolated disagreements and the first positions of consecutive disagreement runs. The resulting onset scores are not calibrated probabilities.

The GNN operated on merged SNV/indel, SV and 5mC records. A window of at most 41 variants (82 allele nodes) was opened around every phased variant with PE ≥ 0.8, extending 20 variants on either side.

Each allele node carried 31 features in four groups: phasing evidence (PE, the genotype and allele indices, the trigger and phase-set indicators, the signed offset from the trigger normalized by the largest absolute offset in the window, the two weighted haplotype totals h_1 and h_2, and six transforms of them); genomic context, derived from the reference sequence and input variant positions (homopolymer length, GC content, tandem-repeat unit count, local sequence entropy, variant density and distance to the nearest indel); graph structure (out-degree, phase-set size, log offset from the trigger, dispersion of the outgoing edge weights and a bridge-vertex indicator); and four one-hot variant-class indicators. Six edge features summarized vote weight, genomic distance, allele matching and phase-set relations. Edges were restricted to the window, sym-

metrized and supplemented with mean-feature self-loops. Exact definitions and normalizations are in Supplementary Tables 1–2.

The network used four GPS graph-transformer layers[37], each combining a local four-head GATv2 branch[38,39] and a global four-head self-attention branch[40], followed by residual normalization and a 128–256–128 feed-forward block. The local GATv2 branch aggregates information along edges of the phasing graph, using edge features to weight the contributions of neighbouring allele nodes. The global self-attention branch allows each node to access information from all nodes within the same window, including those without a direct graph connection. A two-layer classifier produced the two-class softmax output. Architecture and inference details are in Supplementary Method 4, Table 3 and Supplementary Fig. 6.

Only variants no further from the trigger than half the largest trigger distance in the window were scored; the same rule defined the nodes used for training loss. Nodes outside that zone supplied context only. Because the zone is defined on genomic distance and not on variant index, the number of variants it holds depends on the local spacing. The two allele-node scores were averaged, followed by averaging across the windows in which the variant lay inside the scoring zone.

A variant with mean score at least 0.35 was unphased while preserving its genotype. Bridge protection was disabled. If removing these variants disconnected a block, the surviving connected components became separate phase sets (Supplementary Method 4).

Training windows came from the same HG002 nanopore co-phasing runs used for evaluation at 10–20× (six coverages and ten down-sampling replicates, 60 runs). Of 722,116 windows, chr1–14 and chr18–20 supplied training (602,788), chr15–16 validation (67,880), and chr17, chr21 and chr22 testing (51,448); the chromosome sets were disjoint. No test-chromosome window entered gradient updates, batch-normalization statistics or checkpoint selection. Windows contained all four variant classes, but only benchmarked SNV and indel nodes had labels, and SV and 5mC correction relies on the model trained on those labelled small variants. The training and evaluation truth was the v5.0q small-variant benchmark. Training windows were limited to 10–20×, whereas results also included 30–60× and PacBio HiFi data.

Labels described local phase orientation. Within each window and phase set, the first benchmarked variant anchored the orientation; later disagreements were labelled as a single-variant flip or unphase event, or as a break at the first site of a longer run. Benchmark-absent variants and SV/5mC records were masked from the loss but retained as graph context. The released model was trained and applied in binary mode. The binary classifier merged the flip, unphase and break labels into one error class; predicted targets were subsequently unphased. Error labels accounted for 2.96% of labelled central nodes in the training split. Correct nodes were sampled independently in each minibatch at a ratio of at most four per error node, and the resulting cross-entropy used focal weighting[41] with $\gamma = 1.5$ and label smoothing 0.05. Training used haplotype-swap and coverage-scaling augmentations; after a swap, every edge took the vote weight of the allele node it then left. The optimizer, stopping rule and environment are listed in Supplementary Table 4.

We used a normalized net-gain criterion for checkpoint selection, assigning unit benefit to each true-positive detection of a label-defined correction target and unit cost to each false-

positive prediction. Checkpoints were selected on chr15–16 by maximizing

$$S = R\left(2 - \frac{1}{P}\right) \tag{5}$$

over 81 score thresholds from 0.01 to 0.99, subject to $R \geq 0.005$, where P and R are precision and recall for the label-defined error class. This expression applies for $P > 0.5$; otherwise the score was $P - 1$. The criterion serves as a proxy for correction utility (Supplementary Method 4). Checkpoint details and classification metrics for validation and held-out chromosomes are reported in Supplementary Table 4.

Two label definitions were trained, each with five random seeds (42, 43, 44, 1337 and 3407): benchmark variants matched to the calls by position, and matched by position, REF and ALT, which leaves calls whose alleles differ from the benchmark unlabelled. All other settings were identical. Each of the ten models was applied to the 60 HG002 nanopore SNV-only runs at 10–20×, and its deployment threshold B was set on the validation chromosomes to the same cost of correction, 6.35 phased SNVs withheld per removed switch error, that of the model of the earlier release v2.1 at $B = 0.30$. At that cost, the position-matched labels removed 23.0% of switch errors on the held-out chromosomes (mean of five seeds; range 22.4–23.9%) and the REF/ALT-matched labels 21.6% (20.8–22.9%), although the latter had higher held-out classification scores (Supplementary Table 18). The position-matched labels were therefore kept, and the released model is the one that removed the most switch errors on the validation chromosomes (seed 42). The threshold selected on the validation chromosomes was 0.34 and was rounded to $B = 0.35$ for deployment. In the released implementation, it removed 34.3% of switch errors at 7.98 withheld SNVs per removed error over the whole genome, and 23.5% at 9.26 on the held-out chromosomes (Supplementary Method 4 and Supplementary Table 15).

Inference was implemented in C++ without external runtime dependencies; the export procedure and numerical agreement checks are described in Supplementary Method 4.

Haplotagging

Reads were tagged from the phased SNV, indel, SV and 5mC alleles they carried. Each informative observation contributed one vote, and a read was assigned to the majority haplotype when its fraction was at least 0.60; reads whose informative variants belonged to multiple phase sets or lacked a majority were left untagged. Tagged reads received HP, PS and phasing-quality fields. SV alleles were re-derived from the CIGAR and 5mC evidence was matched by read name. Full evidence and quality definitions are in Supplementary Method 5.

Benchmark data

The benchmark used HG002 (NA24385) Oxford Nanopore R10.4.1 reads from the ONT `giab_lsk114_2022.12` release: two PromethION flow cells (PAG65784 and PAG68757), SQK-LSK114, 400 bases per second and 4-kHz sampling, basecalled with Dorado `dna_r10.4.1_e8.2_400bps_sup@v4.0.0` and `5mCG_5hmCG@v2`; their read N50 values were 29.2 and 29.4 kb. Reads were aligned to the GRCh38 no-alt analysis set by Oxford Nanopore with minimap2 v2.24 (`-y -ax map-ont`), retaining reads with mean base quality ≥ 10. The 60× data

set is a random 91.14% subset of the 70.4× alignment (samtools v1.15, `view -s 0.9114`), and the 10–50× data sets were down-sampled from it with samtools v1.23.1[42] `view -s` *seed.fraction*. Ten replicates were used at 10–20× and one at 30–60×; variants were called separately at every coverage and replicate. SNVs and indels came from PEPPER-Margin-DeepVariant r0.8, which reports unphased genotypes, SVs from Sniffles2 v2.8.0, and allele-specific 5mC sites from LongPhase 2. The seeds were 1–9 and 867 at 10× (fraction 0.1666), 12–21 at 12× (0.200), 22–31 at 14× (0.233), 32–41 at 16× (0.266), 42–51 at 18× (0.300), 1–10 at 20× (0.3333), and 42 at 30, 40 and 50× (0.5, 0.6667 and 0.8333). PEPPER-Margin-DeepVariant used the `-ont_r10_q20` model for nanopore data and `-hifi` for HiFi data. HG002 PacBio HiFi (Revio) reads were taken from a 48× GIAB alignment (Supplementary Table 5), which was used as 50×; 10–40× data sets were down-sampled from it with samtools v1.23.1 `view -subsample` (seed 1; fractions 0.2083, 0.4167, 0.6250 and 0.8333), and variants were called separately at every coverage.

Accuracy was scored with `longphase compare` on autosomes 1–22 against two HG002 truth sets: GIAB v4.2.1[29], in the version with phase transferred from a hifiasm assembly (the v4.2.1 benchmark VCF itself is unphased), restricted to biallelic heterozygous records; and GIAB v5.0q[32], derived from the diploid T2T-HG002 assembly[30], in which every heterozygous variant is phased. The truth VCFs were not restricted to their benchmark regions, and a call was scored only where the truth VCF holds the same REF and ALT alleles in a heterozygous genotype. Both truth VCFs list heterozygous variants outside their benchmark regions (9.3% of the autosomal heterozygous records of v5.0q and 3.4% of v4.2.1), so the error counts include sites that the benchmark regions do not cover, among them satellite and segmental-duplication sequence that the T2T-HG002 assembly resolves[30]. To locate switch errors and to score indels, every consecutive variant pair in an intersected block was recorded with a re-implementation of `longphase compare`, which reproduced its counts exactly in all 1,024 scored runs. A pair was assigned to a region when both variants lay in it; regions were the intersection and the differences of the two sets of benchmark regions, and the GIAB v3.6 genome stratifications[25]. Indel switch errors were counted over consecutive pairs that included at least one indel, and the indel Hamming distance was computed in the block orientation chosen on all variants.

Comparison with other methods

WhatsHap v2.8[7] and HapCUT2 v1.3.4[8] were run on the same alignments and variant calls; cross-tool SNV comparisons used SNV-only phasing, with indel phasing disabled in WhatsHap. LongPhase 1.0[11] and LongPhase 2 were run with the nanopore setting, and GNN correction used PE trigger 0.8, score threshold 0.35, no bridge protection and block splitting. Co-phasing experiments separately evaluated the seven variant-class combinations listed in Supplementary Table 10, with and without correction. Indel co-phasing was also compared with WhatsHap's default indel phasing. Margin v2.3.1[9] and Ralphi[27] were evaluated at 10–20×, with ten down-sampling replicates per coverage; these tools were not evaluated at 30–60×. Their configurations and that of MethPhaser[21] are listed in Supplementary Table 5. MethPhaser was given the SNV-only output of LongPhase 2 without GNN correction instead of the WhatsHap or HapCUT2 phasing its authors recommend as input. MethPhaser and the joint SNV and 5mC phasing of LongPhase 2 started from the same SNV phasing. Without GNN correction, differences between

them reflect how methylation is used, after phasing to bridge existing blocks (MethPhaser) or during phasing as additional graph evidence (LongPhase 2); with GNN correction, they also include its effect. GCphase[43] was excluded from the genome-wide comparison because it ignored the FILTER field and phased homozygous-reference calls as heterozygous variants when given the full input VCF (about 1.2 million such calls in the 10× replicate-1 output). GCphase was not evaluated with input restricted to heterozygous SNVs. The HiFi call sets were phased SNV-only with LongPhase 2 (`-pb`), with and without correction, with WhatsHap, with HapCUT2 and with HiPhase (see below); the GNN weights were trained on nanopore data. Four further phasers were run on replicate 1, at 10–60× in steps of 10× on nanopore data and at 10–50× on HiFi data, with default parameters or the workflow their documentation describes (Supplementary Table 5). On nanopore data, HapBridge[22] was given the same input as MethPhaser, and LongHap[23] phased SNVs with and without 5mC and, from the same reads, SNVs, indels and SVs with 5mC; because LongHap requires sequence-resolved alleles, Sniffles2 deletions and insertions were given to it with their allele sequences, and inversions, duplications and breakends were omitted. TinkerHap[34] was run on both platforms and HiPhase[33] on HiFi data. Both phase indels by default and have no SNV-only mode, so each was run on the full call sets and on the call sets restricted to SNVs, the variants that WhatsHap phases with `-only-snvs`. For the comparison with indels, LongPhase 2, WhatsHap and HapCUT2 were also run with indel phasing on HiFi data, and HapCUT2 on nanopore data. These runs were scored in the same way as all others (Supplementary Table 18). WhatsHap was run as `whatshap phase -ignore-read-groups -reference` with `-only-snvs` for SNV comparisons, and HapCUT2 as `extractHAIRS -ont 1` (`-pacbio 1` for HiFi) followed by `HAPCUT2` with default parameters, including its default likelihood-based pruning, on the PASS heterozygous SNVs. HapCUT2 rejects genotypes other than two alleles from 0, 1 and 2, so the HiFi call sets given to it omitted such records (134–204 uncalled reference candidates per coverage and one multi-allelic indel at 50×); none of them was a heterozygous SNV. LongPhase 1.0 was re-run on every down-sampling replicate. All tools ran on the same workstation.

Evaluation metrics

Metrics (Supplementary Fig. 8) were computed by intersecting heterozygous query and truth records by position and REF/ALT alleles, then partitioning the intersection into blocks phased in both files; singleton blocks were ignored. Within each block, switch errors were the Hamming distance between truth and query switch-encoded haplotype strings, and the switch-error rate divided their number by the assessed consecutive SNV pairs. Isolated switches and two-consecutive-switch flips were reported separately. Hamming distance was the block-wise minimum over the two orientations, divided by the number of compared variants; unlike switch rate, it included indels when present in both files, so it is not comparable between configurations with and without indels, which we therefore compared by SNV switch error rate. Because a longer block can contain a longer segment of wrong orientation, block-wise Hamming distance tends to increase with block length, so it does not compare tools with different block lengths on equal terms. Lower Hamming distances were also observed when LongPhase 2 produced longer blocks than WhatsHap. In SNV-only phasing at 30–60×, LongPhase 2 had 2–5% longer blocks

than WhatsHap and 1.4–1.8-fold lower Hamming distances. In joint SNV and indel phasing, the blocks of WhatsHap were 12–33% longer, but its indel Hamming distance was 1.8–2.7-fold higher, and this difference increased with coverage as the difference in block length decreased. MethPhaser lengthens its input blocks by joining them without changing phase within them; these joins increased the Hamming distance 1.8–2.8-fold over that of its input at 20–60×. Phased fraction was the number of truth heterozygous SNVs or indels that were phased by the query divided by all truth heterozygous variants of that class. Block N50 was the span-weighted median of query phase-set spans of at least two variants, pooled over chromosomes. Switch-error positions were exported for the unphased-SNV analysis. The exact block and denominator definitions are in Supplementary Method 6.

SV phase evaluation

SV phase was scored on chr1–22 against the variants of the T2T-HG002 Q100 v1.1 assembly called on GRCh38 by dipcall[30,44], which assign every variant, including SVs, to a haplotype; records with a missing allele on either haplotype were excluded. SVs were called with Sniffles2 v2.8.0 with the reference genome and supporting read names on both platforms. A phased heterozygous deletion or insertion of at least 50 bp matched a truth SV of the same type with a size similarity of at least 70% and a position within 1 kb or, for deletions, a reciprocal overlap of at least 50%. Each block was oriented by the majority of its SNVs, and a matched SV was correct when its oriented phase agreed with the assembly. The SV error rate was the number of incorrectly phased SVs divided by the number of correctly and incorrectly phased SVs. WhatsHap v2.8 (`-indels`) and HiPhase v1.7.0 read SVs only as sequence-resolved records in the variant VCF; for these tools, deletions were therefore given the reference sequence as REF and insertions the inserted sequence as ALT, and both were merged with the SNV (and indel) calls. LongPhase 2 received the unmodified SV calls. Inversions, duplications, breakends, symbolic insertions and records with an unknown reference base (296–580 per run) could not be converted; they were not given to WhatsHap and HiPhase and were not scored for any tool.

Analysis of unphased SNVs

An SNV was counted as unphased by GNN correction when it was phased in the pre-correction output and unphased afterwards. Such sites were classified against the v5.0q benchmark VCF, matched by chromosome and position and without a benchmark BED, as absent, homozygous or heterozygous (Fig. 5f); the heterozygous count was also reported with matching REF and ALT alleles, the rule used by `compare`. The analysis used the 10–60× outputs of replicate 1 with the final GNN model.

At 60×, SNVs unphased by GNN correction were also classified against the T2T-HG002 assembly projected to GRCh38 as outside the 1:1 assembly BED, inside without an assembly variant at the same start coordinate, or inside with one; alleles were not compared. All chromosomes, including chrX and chrY, were analysed, and autosome-only values were derived from the per-chromosome counts. A random sample of 50,000 SNVs phased before correction, which included sites that GNN correction later unphased, provided the background; enrichment was summarized by chromosome and 1-Mb bins.

To compare LongPhase 2 and WhatsHap, the heterozygous SNV calls with single-base alleles and genotype 0/1, which were identical in the two output VCFs, were grouped by the tool that phased them: both tools, LongPhase 2 only, WhatsHap only or neither; calls phased by WhatsHap only were further split by whether LongPhase 2 had already left them unphased before GNN correction. Each call was classified against the v5.0q chr1–22 VCF, without a benchmark BED, as heterozygous with the same position and alleles, heterozygous with other alleles, homozygous or absent; calls on chrX and chrY were counted separately. The calls phased by a tool were those phased by both tools plus those phased by that tool only; phased calls at positions with a homozygous v5.0q record were counted as genotype errors (Fig. 5e). For every SNV-only run of both tools, and on HiFi data also of HiPhase, the share of the phased heterozygous SNV calls on chr1–22 that matched a benchmark heterozygous SNV (precision) was computed from the `compare` count of matched calls, and the F1 score as the harmonic mean of precision and the phased fraction. For the matched calls phased by one tool only, the phase assigned by that tool was scored as in `compare`, and a call was counted as wrongly phased when its phase was opposite to the majority orientation of its block relative to the truth. A switch-error interval extends from the first SNV of a switch-error pair to the base before the second; the share of the calls phased by one tool only inside that tool's intervals was compared with the share of all heterozygous SNV calls phased by that tool. Calls were also annotated by read depth, by the v5.0q benchmark regions, by GIAB v3.6 segmental duplications and by the one-to-one dipcall alignment (Supplementary Data 1). [Add the script behind Fig. 5f (`analyze_unphased.py`) to the repository; the scripts for Fig. 5b,d,e are in `notes/unphased`.]

Runtime and memory

Wall-clock time, CPU time and peak resident memory were measured with `/usr/bin/time`; reported LongPhase 2 times included alignment reading, graph construction, phasing, GNN correction and VCF writing, and excluded alignment, variant calling and down-sampling. All runs used one workstation (AMD Ryzen 9 9950X3D, 16 cores and 32 threads, 249 GB of memory, Ubuntu 24.04, input and output on a hard-disk drive). LongPhase 2 was compiled with g++ 13.3 at `-O3`. Times were taken once per coverage, on replicate 1 at 10–20× and on the single data sets at 30–60×. WhatsHap and HapCUT2 ran on one thread and were timed in the runs that produced the phased VCFs. LongPhase 2 was timed with 24 threads at every coverage and on one thread at 10 and 60×, in separate runs made one at a time with no other jobs running; each produced output identical to the reported VCF. The logs of the HapCUT2 runs at 30 and 40× were not retained, so these two runs were repeated with the same commands and inputs, one at a time with no other jobs running; their phased output was identical to that of the original runs. On HiFi data (10–50×, one run per coverage), LongPhase 2 was timed in the same way with 24 threads, and HapCUT2 in the runs that produced its output. The WhatsHap runs at 10 and 20× had shared the workstation with other jobs; they were therefore repeated one at a time with no other jobs running, each after reading its alignment file once, and each produced output identical to the original run. WhatsHap and HapCUT2 have no thread option; running them on each chromosome in parallel would shorten their wall-clock time but not their CPU time, so CPU time is reported next to wall-clock time (Supplementary Fig. 10 and Supplementary

Table 16).

Statistics and reproducibility

Ten nanopore down-sampling replicates at 10–20× were summarized by their mean and standard deviation, while 30–60× and HiFi used one down-sampling per coverage. Subsamples for each platform were drawn from the same source sequencing data set of HG002. The reported standard deviations therefore describe variation due to read subsampling, conditional on that data set, rather than variation among independent sequencing experiments. Subsampling variability was not estimated at 30–60× or for HiFi. Tools compared at each coverage and replicate used the same sampled reads, so fold differences between tools and the effect of GNN correction at 10–20× were also computed within each replicate (Supplementary Table 8). No statistical tests, sample-size calculation, randomization or blinding were used. GNN inference used fixed model weights and no sampling.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data analysed in this study are publicly available: the GRCh38 no-alt analysis set from NCBI (`https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/`); the GIAB HG002 v4.2.1 phase-transfer VCF (`https://ftp-trace.ncbi.nlm.nih.gov/giab/ftp/release/AshkenazimTrio/HG002_NA24385_son/NISTv4.2.1/GRCh38/SupplementaryFiles/`) and the v5.0q benchmark (`https://ftp-trace.ncbi.nlm.nih.gov/giab/ftp/release/AshkenazimTrio/HG002_NA24385_son/v5.0q/`) from the GIAB FTP site; and the HG002 nanopore R10.4.1 reads and alignments from the ONT open-data archive (`s3://ont-open-data/giab_lsk114_2022.12/`) under a CC BY-NC 4.0 licence. The truth files used are listed in Supplementary Table 5.

Code availability

LongPhase 2 is open source under the GNU General Public License v3.0 at `https://github.com/twolinin/longphase`; the repository includes a demonstration data set and workflow. The reported results correspond to release v2.1.2 (commit `6b8f2dd5050028a9546026a3311a7c52bc5389db`). The GNN training code, the trained model and the scripts used for the analyses and figures are available in the same repository.[Add the training, analysis and figure-generation scripts to the public repository under an OSI-approved licence, with development-machine paths parameterized, before submission.]

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